

Week 4 HW Question 7:

Date:

$$A = \begin{bmatrix} 2 & -2 & -1 \\ 1 & 3 & 2 \\ -8 & 8 & 4 \end{bmatrix}$$

step 1: $4R_1 + R_3 \rightarrow R_3$

$$\begin{bmatrix} 2 & -2 & -1 \\ 1 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

step 2: Any matrix with an entire row of zeros has a determinant of 0.

Therefore, $\det(A) = 0$.